4. THE NORMAL STATES OF ATOMIC NUCLEI

I. THE BINDING ENERGY OF THE NUCLEI

Of the questions mentioned at the conclusion of the previous lecture, let us discuss first the one concerning the forces which are operative between the building blocks of the nucleus and hold these particles together. This question may be formulated as follows: What physical magnitude or what property of the atom determines its stability? One might be inclined to think, at first, that this question is a difficult one, and that in order to answer it, we must be familiar with the entire mechanical system represented by an atomic nucleus, in all its details. However, fortunately, this is not the case. There are a few fundamental laws which enable us to discuss the stability and other general properties of a system even though we may be ignorant of the nature of the forces acting within it, as well as of the details of its structure. These are the well known laws of the conservation of mass, energy, etc., which state that these things can neither be created out of nothing nor be annihilated. Those laws which are of primary significance here, are the laws of the conservation of energy, of electric charge, and of angular momentum.

We shall begin with the law of the conservation of energy. Let us assume that it is possible to employ certain forces so as actually to remove a particle from a nucleus (which latter consists of protons and neutrons). We may make this theoretically possible by the naive assumption that we can actually seize this particle and transport it to a point far away from the nucleus. Since the particle was originally firmly bound to the nucleus, it is now attracted by it, so that a certain amount of work is required in order to remove it; in other words, energy must be introduced into the system. Now, according to the energy laws, this work, this imported energy, is quite independent of the method employed to remove the particle. Therefore it follows that every particle is bound within the nucleus by a definite quantity of energy, and this quantity of energy can be calculated if it is somehow possible to ascertain the quantity of energy of the system before and after the removal of the particle. Let us now proceed to define the concept of the *binding energy* of a nucleus. This term stands for that change in nuclear energy which takes place when the constituent parts of the nucleus—originally far away from each other—are combined to form that nucleus. Since the reverse of this process, the breaking up of the nucleus, requires an expenditure of energy—in other words, energy must be applied to the nucleus—the nucleus most lose energy in the process of its formation. Therefore, the binding energy of an atomic nucleus is always